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A1S2 Revision

Statistics and Data Science (Universiteit Stellenbosch)



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Stats: Revision

Chapter 7

Mean: $\mu = \frac{\sum X_i}{N}$ — N: Population

Standard Deviation: $\sigma = \sqrt{\frac{\sum (X_i - \mu)^2}{N}}$

Standard Error of the Mean: $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$

If population is Normal, sampling distribution of $\bar{X}$ is normally distributed with: $\mu_{\bar{x}} = \mu$ / $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$

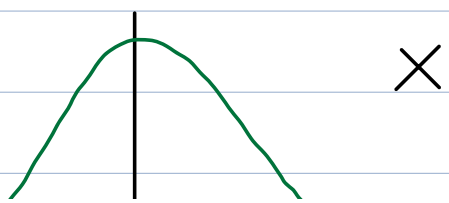
Z-value for sampling distribution of $\bar{X}$:

$$z = \frac{(\bar{x} - \mu_{\bar{x}})}{\sigma_{\bar{x}}} = \frac{(\bar{x} - \mu)}{\frac{\sigma}{\sqrt{n}}}$$

$\bar{x}$: Sample mean; μ: Population mean; σ: Population standard deviation; n: Sample size

$\mu_{\bar{x}} = \mu$

($\bar{X}$ is unbiased)

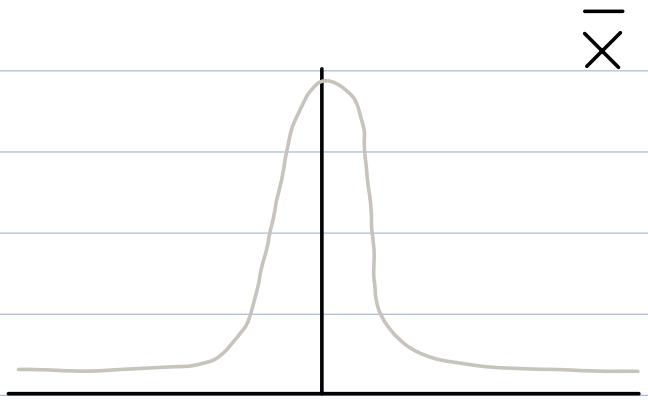


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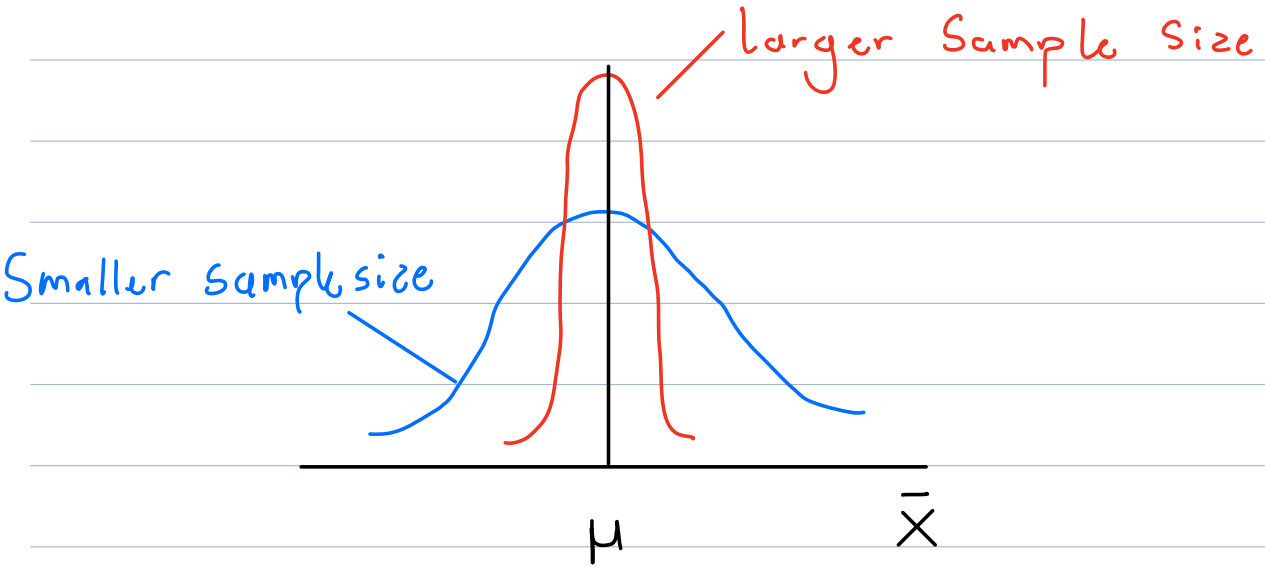
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Normal Population Distribution



Normal Sampling Distribution
(same mean)



Lower / upper limit of the interval

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$$\bar{x}_L = \mu + Z\frac{\sigma}{\sqrt{n}}$$

$$\bar{x}_U = \mu + Z\frac{\sigma}{\sqrt{n}}$$

Central Limit Theorem :

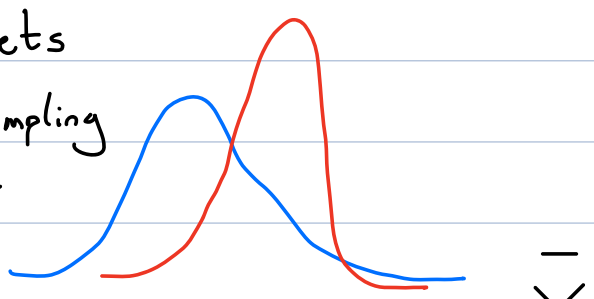
* Even when population is not normal, sample means from population will be approximately normal as long as sample size is large enough.

Properties of sampling distibution :

$$\mu_{\bar{x}} = \mu$$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

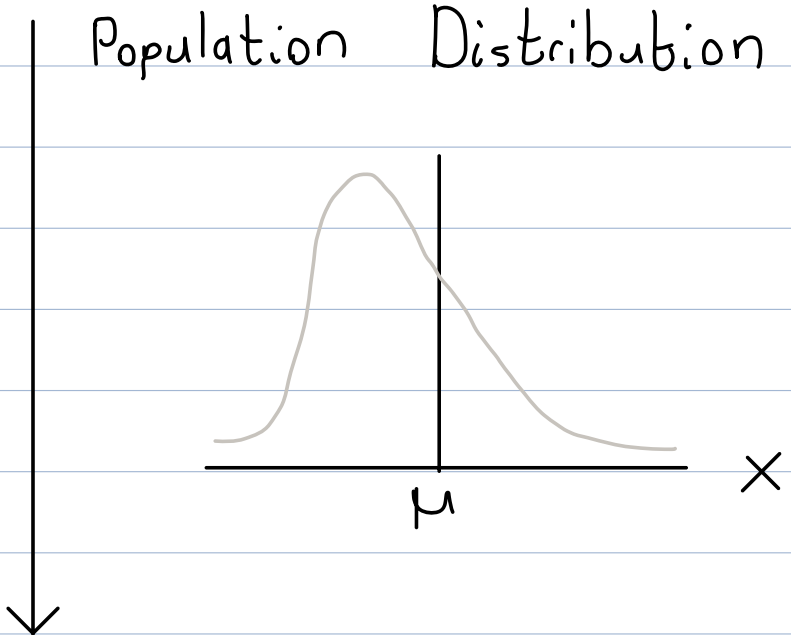
Sample size gets large enough, sampling distribution of sample becomes almost normal regardless of shape



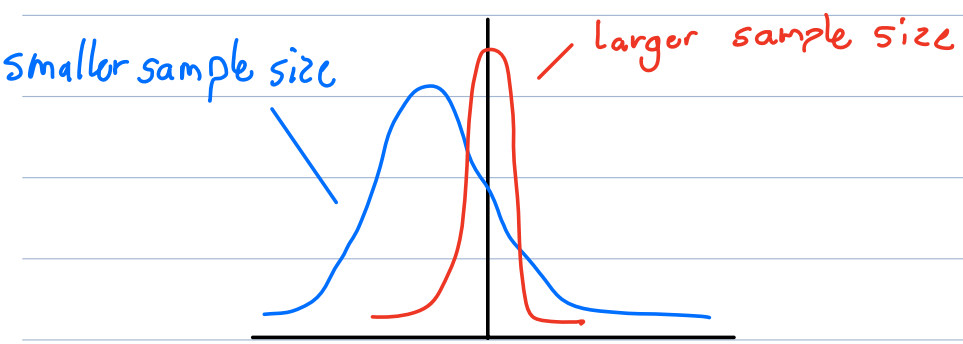
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Sample mean sampling distribution, if population is not normal:



Sampling Distribution
(becomes normal as n increases)



How large is large enough?

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$n > 30$ will give a sampling distribution that is nearly normal.

$n > 15$ is large enough for fairly symmetric distributions

Population Proportions

π = proportion of population having specific characteristic

Sample proportion (p) $\longrightarrow$ provides estimate of π

$$p = \frac{X}{n} = \frac{\text{no. of items in sample with characteristic}}{\text{sample size}}$$

$$0 \leq p \leq 1$$

p is approximately distributed as a normal distribution, when n is large.

Sampling Distribution of p

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$n \cdot \pi \geq 5$ and $n(1-\pi) \geq 5$

where

$$\mu_p = \pi \quad \text{and} \quad \sigma_p = \sqrt{\frac{\pi(1-\pi)}{n}}$$

π = population proportion

Z-value for proportions

To standardize p to a Z-value:

$$Z = \frac{p - \pi}{\sigma_p} = \frac{p - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}}$$

Sampling distribution of the difference between 2 sample means

Assumptions: * samples drawn randomly + independently

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* Pop. distributions are normal / or both sample sizes are large ($n \geq 30$)
* Population standard deviations known

sampling distribution : $\bar{X}_1 - \bar{X}_2$

$$\mu_{\bar{x}_1 - \bar{x}_2} = \mu_1 - \mu_2$$

standard error :

$$\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

z - value :

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{x}_1 - \bar{x}_2}}$$

General note on utilising symmetry

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Given symmetrical nature of normal distribution:

$$p(Z > z) = 1 - p(Z \leq z) = P(Z < -z)$$

Region symmetrical around the mean:

$$P(z_1 < Z < z_2) = P(Z \leq z_2) - P(Z \leq z_1) = 1 - 2P(Z < z_1)$$

Sampling from Finite Populations

when sampling without replacement from finite populations, where sample size (n) is more than 5% of population size ($\frac{n}{N} > 0,05$)

↓

Finite Population Correction Factor is used to calculate $\sigma_{\bar{x}} / \sigma_p$ (fpc)

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$$fpc = \sqrt{\frac{N-n}{N-1}}$$

standard error of mean :

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

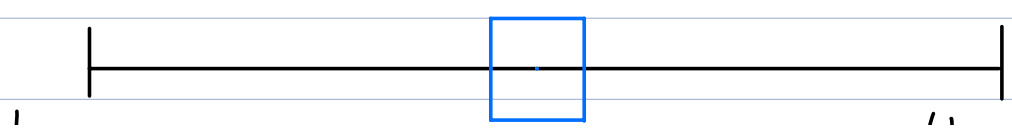
standard error of proportion :

$$\sigma_p = \sqrt{\frac{\pi(1-\pi)}{n}} \sqrt{\frac{N-n}{N-1}}$$

Chapter 8

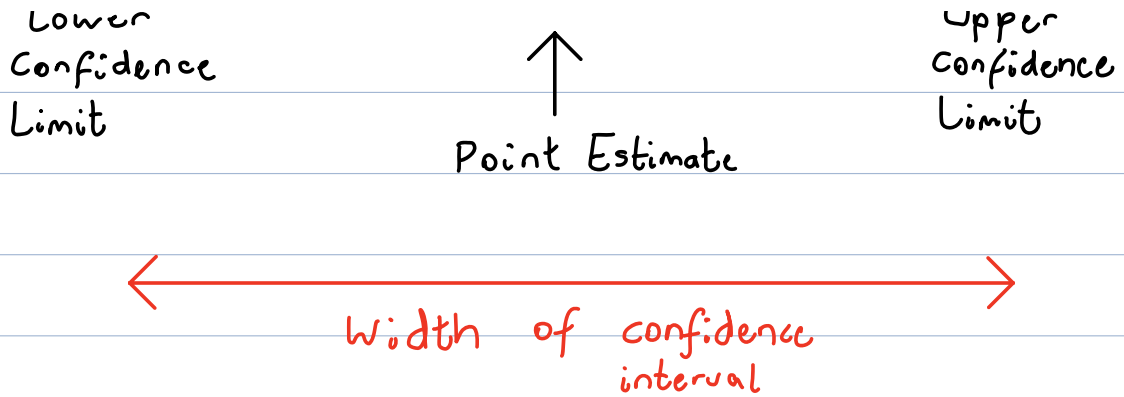
Point + Interval estimates

* Point estimate is single number
* Confidence interval provides additional info about precision of estimate



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Estimate a Population Parameter		with sample statistic
Mean	μ	$\bar{X}$
Proportion	π	p

Interval estimate provides more info about a population characteristic than a point estimate

called confidence intervals

5% / 2 = 2,5% $Z(0,025)$

95% confidence is based on $Z = 1,96$

General Formula for confidence intervals :

Point estimate $\pm$ (Critical value)(Standard error)

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sample statistic estimating population parameter of interest

table value based on sampling distribution of point estimate + desired confidence level

standard deviation of the point estimate

Confidence level

* Confidence the interval will contain the unknown population parameter (Shown as percentage)

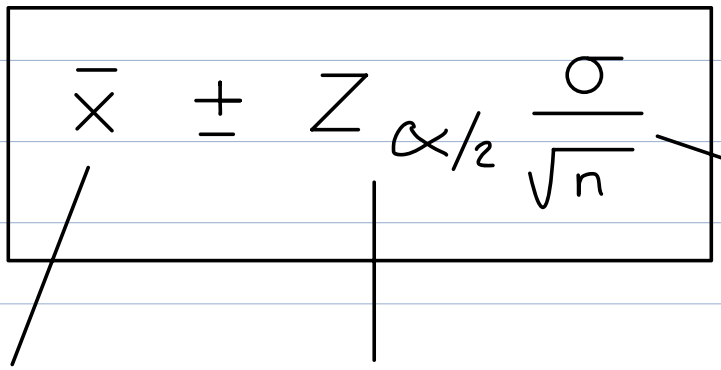
$(1 - \alpha)$

Confidence Interval for μ (σ known)

Assumptions:
* Population σ is known
* Population is normally distributed
* If population is not normal, use large sample ($n > 30$)

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Common levels of confidence

Confidence Level	Confidence Coefficient $(1-\alpha)$	$Z_{\alpha/2}$ value
80 %	0,80	1,282
90 %	0,90	1,645
95 %	0,95	1,960
98 %	0,98	2,326
99 %	0,99	2,576
99,8 %	0,998	3,090
99,9 %	0,999	3,291

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Confidence Interval for μ (σ unknown)

* If σ is unknown, substitute sample standard deviation, S
  - t distribution is used instead of normal distribution

Assumptions:

* Population σ is unknown
* Population is normally distributed

Confidence Interval estimate:

$$\bar{X} \pm t_{\alpha/2} \frac{S}{\sqrt{n}}$$

(where $t_{\alpha/2}$ is the critical value of the t distribution with $n-1$ degrees of freedom + an area of $\alpha/2$ in each tail)

* t is a family of distributions

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$t_{\alpha/2}$ depends on degrees of freedom (df)

$$df = n - 1$$

Confidence intervals for π (Population Proportion)

$$\sigma_p = \sqrt{\frac{\pi(1-\pi)}{n}}$$

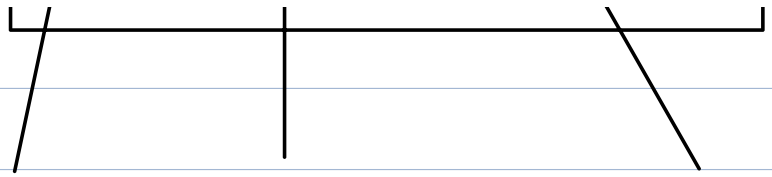
Sample data estimated with:

$$\sqrt{\frac{p(1-p)}{n}}$$

upper/lower confidence limits for population proportion:

$$p \pm Z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}$$

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sample proportion

standard normal value for the level of confidence desired

sample size

Error of estimation :

* error of estimation defined as difference between estimator + parameter

margin of error (e)

(Sampling error)

$$e = Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$n = \frac{Z^2_{\alpha/2}\, \sigma^2}{e^2}$$

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(For the proportion)

$$e = Z\sqrt{\frac{\pi(1-\pi)}{n}}$$

$$n = \frac{Z^2_{\alpha/2}\,\pi(1-\pi)}{e^2}$$

Interval and sample size estimation for finite populations

Confidence interval for μ with a fpc

$$\bar{x} \pm t_{\alpha/2}\,\frac{S}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}$$

$$p \pm Z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}\sqrt{\frac{N-n}{N-1}}$$

fpc reduces standard error of sample mean or sample proportion

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Calculating sample size (n_0) without fpc

For μ : $n_0 = \dfrac{Z^2_{\alpha/2}\,\sigma^2}{e^2}$

For π : $n_0 = \dfrac{Z^2_{\alpha/2}\,\pi(1-\pi)}{e^2}$

Apply fpc to arrive at final sample size (n)

$$n = n_0 N \,/\, (n_0 + (N-1))$$

Bootstrapping

Bootstrapping Is A Method To Use When Population Is Not Normal

DCOVA

To estimate a population mean using bootstrapping, you would:

1. Select a random sample of size n without replacement from a population of size N.
2. Resample the initial sample by selecting n values with replacement from the initial sample.
3. Compute $\overline{X}$ from this resample.
4. Repeat steps 2 & 3 m different times.
5. Construct the resampling distribution of $\overline{X}$.
6. Construct an ordered array of the entire set of resampled $\overline{X}$'s.
7. In this ordered array find the value that cuts off the smallest α/2(100%) and the value that cuts off the largest α/2(100%). These values provide the lower and upper limits of the bootstrap confidence interval estimate of μ.

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Chapter 9

Hypothesis testing: (σ known)

1) Straightfoward claim about population mean, referred to as null hypothesis (H_0)

2) Mutually exclusive claim to null hypothesis, referred to as alternative hypothesis (H_1)

3) Begin with assumption that H_0 is true

4) Examine sample statistics to see if it better supports H_0 / H_1

Type I Error: * Reject a true H_0 (true)
* "False alarm"
* Probability of Type I error is denoted by α (increases)

Type II Error: * Failure to reject a false H_0 (false)

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* "Missed opportunity"
* Probability of Type II error is denoted by β (decreases)

Z test of Hypothesis for Mean (σ known)

$$z_{stat} = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Hypothesis testing: (σ unknown)

Z distribution ⟶ t distribution

Assume that distribution is normal

$$t_{stat} = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

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Hypothesis Tests for proportions

$$p = \frac{x}{n} = \frac{\text{no. in category of interest within sample}}{\text{sample size}}$$

when:

$$n\pi \text{ and } n(1-\pi) \geq 5$$

then:

$$\mu = \pi \qquad \sigma = \sqrt{\frac{\pi(1-\pi)}{n}}$$

$$z\,\text{stat} = \frac{p - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}}$$

$\therefore$ since $x = np$, z stat can be re-expressed as:

$$z\,\text{stat} = \frac{x - n\pi}{\sqrt{n\pi(1-\pi)}}$$

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Power of Hypothesis Test:

$(1 - \beta) \longrightarrow$ probability of rejecting a false H_0

β = probability of Type II error (failing to reject a false H_0)

$$Z\ stat = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Hypothesis for proportions

$$\frac{p - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}}$$

Two-tail:

$$t\ stat = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

Two Population:

$$t\ stat = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

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Related samples:

$$t_{stat} = \frac{\bar{d} - \mu_d}{\frac{s_d}{\sqrt{n}}}$$

Two population:

$$z_{stat} = \frac{(p_1 - p_2) - (\pi_1 - \pi_2)}{\sqrt{\bar{p}(1-\bar{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Pooled estimate: $\frac{x_1 + x_2}{n_1 + n_2}$

$$F_{stat} = \frac{s_1^2}{s_2^2}$$

$$P(\bar{x} > 95$$

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$$\sigma_p = \sqrt{\frac{\pi(1-\pi)}{n}}$$

$$= \sqrt{\frac{0,7(1-0,7)}{95}}$$

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